

Lab 05

Predicting environmental outcomes

Teaching Week 05 | Individual work. Submit in R or Python.

Purpose

This lab introduces predictive evidence through an environmental regression task. You will define a target, protect the evaluation split, compare a fitted model with a simple baseline, and interpret errors in the target's original units.

Learning outcomes

After completing this lab, you should be able to

- distinguish prediction from description and explanation;
- construct a leakage-aware training and evaluation split;
- compare a model with a meaningful baseline using several metrics; and
- diagnose where prediction errors are largest.

Dataset choice

Choose one of the verified open datasets below. Each option supports the assessment when your analytical table satisfies the data contract in the next section. The expected variables describe the published record and may differ in detail from the extract you retrieve; reconcile any difference in your report.

Option A | Halifax County Water Quality Data

Publisher. Nova Scotia Environment and Climate Change

Official record. <https://data.novascotia.ca/d/x9dy-aa19>

Record identifier. x9dy-aa19

Licence and format. Nova Scotia Open Government Licence; CSV

Expected variables. waterbody; station; lease; timestamp; latitude; longitude; temperature; salinity; dissolved oxygen; sensor depths; quality flags

Use in this lab. predict a selected environmental measurement

Why it fits. Repeated numeric sensor measurements support baselines, train-test evaluation, and error metrics.

Option B | Fuel Consumption Ratings

Publisher. Natural Resources Canada

Official record. <https://open.canada.ca/data/en/dataset/98f1a129-f628-4ce4-b24d-6f16bf24dd64>

Record identifier. 98f1a129-f628-4ce4-b24d-6f16bf24dd64

Licence and format. Open Government Licence - Canada; CSV

Expected variables. model year; make; vehicle class; engine size; cylinders; fuel type; fuel consumption; CO2 emissions

Use in this lab. predict fuel consumption or carbon dioxide emissions

Why it fits. Canadian energy-efficiency and emissions records support leakage-aware splits, baselines, and errors in meaningful units.

Option C | EnerGuide Appliance Energy Consumption

Publisher. Natural Resources Canada

Official record. <https://open.canada.ca/data/en/dataset/fbdfd946-8dd1-4830-a5c9-f8a72d8fabda>

Record identifier. fbdfd946-8dd1-4830-a5c9-f8a72d8fabda

Licence and format. Open Government Licence - Canada; ZIP archives containing CSV files

Expected variables. appliance type; brand; model; capacity; energy use

Use in this lab. predict energy consumption from product attributes

Why it fits. Numeric and categorical predictors support baselines and comparative metrics.

Data contract

Choose one listed dataset and define one numeric prediction target.

- Retain at least 200 complete observations after documented cleaning.
- Include one target and at least three candidate predictors.
- Identify any time, station, site, product, or group column that constrains the split.
- Compare a simple baseline with one fitted predictive model.
- Report at least two appropriate error metrics and interpret their units.
- Fit preprocessing on training data only.
- Reuse prior summaries, cleaning, graphics, and reproducible reporting.
- Document the official record URL, publisher, licence, retrieval steps, and access date.

What the contract does not settle

The contract constrains the table. It does not choose the target, the model, or the split.

Selection checklist

Resolve every unchecked item before you begin the analysis.

- ☐ I chose one option from the dataset catalogue above.
- ☐ I recorded the official URL, publisher, licence, access date, and retrieval steps.
- ☐ I defined one numeric target and at least three predictors.
- ☐ My table satisfies every rule in the data contract.
- ☐ I identified group or time structure that constrains the split.
- ☐ I selected a simple baseline and two interpretable error metrics.
- ☐ I will fit preprocessing on training data only.
- ☐ I will cite the dataset in the report.

Cumulative method

This lab carries forward the earlier data audit, cleaning pipeline, statistical graphics, and reproducible report structure. Your report must show how those earlier decisions support the current analysis rather than repeating them as disconnected exercises.

Work program

1. Define the prediction problem

State the target, prediction unit, available-at-prediction-time features, use case, and one source of leakage.

2. Establish the split and baseline

Reserve evaluation data before fitting transformations. Compare against a defensible mean, median, seasonal, station, or category baseline.

3. Fit one predictive model

Document the representation, fitted model, training-only preprocessing, and seed. Do not tune on the final evaluation data.

4. Evaluate errors

Report at least two appropriate metrics, a predicted-versus-observed view, and error summaries for one meaningful subgroup or range.

5. State the claim

Explain what performance would mean in use, where the model fails, and why predictive association is not a causal explanation.

Reproducible workflow

You may complete and submit the lab in either R or Python. Both paths are marked against the same evidence and rubric, and both must run end to end from one documented entry point in your portfolio repository.

Write your own analysis using the methods taught in the course. Document how to run it from the retained source data and produce every result cited in your report. Choose and justify the variables, preparation steps, and analytical methods required by your question and this lab's data contract. Record the software environment and any randomness that affects reproducibility.

Downloaded data and generated results are not course source, so they never go into the course repository. Commit the report, its source, and the generated tables and figures it cites to your private work repository. Keep that evidence private until the deadline, and publish it in your public GitHub portfolio repository only when the instructor opens sharing. Link rather than redistribute source data unless its licence and size permit it.

Submission package

Keep the lab private before its submission deadline, including cloud notebooks and answer-revealing links. A branch in a public repository is not private. Give the instructor access to your private work repository. First commit the completed report, source, and supporting evidence. Record that commit's full SHA and commit URL in `labs/lab-05/README.md`, then commit this README update separately. A commit cannot contain its own SHA. Push both commits to the default branch before 16:05 Halifax time on October 15, 2026, seven calendar days after the integrated lab. The deadline holds even when no class meets that day. The marker reads that README on the default branch and assesses the snapshot it names. The immutable commit or release URL must resolve to that exact version of `labs/lab-05/`. A branch URL or movable tag alone is not a fixed submission. Include the following:

- a concise PDF report that answers the analytical question, together with the completed selection checklist;
- the complete R or Python source plus the generated tables, figures, and any model metrics referenced in the report;
- a README carrying the submitted commit SHA and its commit URL, the official data URL, retrieval instructions, exact execution command, and environment; and
- no redistributed source data unless its licence and size permit it.

After the deadline and when the instructor opens sharing, publish the same submitted commit and its history in your public GitHub portfolio repository and present that version at the next class meeting, if one follows. Preserve the submitted SHA when merging. Commit regularly as you work. Do not amend, rebase, squash, delete, replace, force-push, or backdate recorded assessment history. Keep submitted commits reachable and make corrections in new commits. The pushed README record and the instructor-retained snapshot provide the independent assessment reference; Git author dates alone are not proof of timely submission.

Assessment standard

The course uses one rubric across all integrated labs. The best ten of the twelve weekly lab results contribute 15% of the course grade.

Table 1. Integrated lab marking criteria.

Criterion	Weight (%)
Dataset provenance and suitability	15
Reuse of earlier methods	20
Current-week analytical method	35
Interpretation and limitations	20
Reproducibility and communication	10
Total	100

Academic integrity, private GitHub exceptions, private grading, and publication limits follow the course syllabus. Generative AI output is prohibited on integrated labs, so generated code, prose, analysis, or figures cannot be submitted for lab credit. Cite the dataset and every non-trivial external source; the submitted analysis must be your own work.